<table>
<tr><th>Course Code</th><th colspan="5">Name of the Course</th></tr>
<tr><td>216U03C405</td><td colspan="5">Electromagnetic Field Theory</td></tr>
<tr><td colspan="6"></td></tr>
<tr><td rowspan="2">Teaching Scheme</td><td>TH</td><td>P</td><td colspan="2">TUT*</td><td>Total</td></tr>
<tr><td>03</td><td>--</td><td colspan="2">01</td><td>04</td></tr>
<tr><td>Credits Assigned</td><td>03</td><td>--</td><td colspan="2">01</td><td>04</td></tr>
<tr><td colspan="6"></td></tr>
<tr><td rowspan="4">Evaluation Scheme</td><td colspan="5">Marks</td></tr>
<tr><td rowspan="2">LAB/TUT CA</td><td colspan="2">CA (TH)</td><td rowspan="2">ESE</td><td rowspan="2">Total</td></tr>
<tr><td>IA</td><td>ISE</td></tr>
<tr><td>25</td><td>20</td><td>30</td><td>50</td><td>125</td></tr>
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* Batch wise Tutorial

Course pre-requisites:

- Differentiation Methods
- Vector Algebra, Vector calculus
- Basic laws of electricity.

Course Objectives:

Electromagnetics principles find applications in various allied disciplines of electronics and telecommunication engineering. This course will help the students to learn basic concepts of electromagnetics, related laws, wave equations and applications.

Course Outcomes:

At the end of successful completion of the course the student will be able to

CO1. Apply basic laws of electrostatics in vector form to find electric fields in materials.

CO2. Apply basic laws of magnetostatics in vector form to find magnetic forces in materials.

CO3. Evaluate parameters of dielectric and conducting media by solving wave equations and study polarization of waves.

CO4. Calculate energy transported by means of electromagnetic waves and reflections of plane waves.

CO5. Apply concepts of electromagnetics to real life applications in communications.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1	**Electrostatics**		10	CO 1
	1.1	**Electric Field:** Coulomb's law, electric field intensity, charge distributions, electric flux density and Gauss's law		
	1.2	**Electric Potential**: Relation between electric field and potential, electric dipole, electric flux and energy density in electrostatic fields		
	1.3	**Electrostatic field in material:** properties of materials, convection and conduction currents, polarization in dielectrics and dielectric constant.		
	1.4	**Electrostatic boundary:** Boundary conditions, Poisson's and Laplace's equation, resistance and capacitance		
2	**Magnetostatics**		10	CO 2
	2.1	**Magnetic Field:** Biot – Savart's law, Ampere's circuit law, magnetic flux density and magnetic field intensity.		
	2.2	**Energy and Potential:** Magnetic scalar and vector potential, energy in magnetic field, magnetic circuits		
	2.3	**Magnetic Force:** Forces due to magnetic fields, Lorentz force equation.		
	2.4	**Magnetic boundary:** Magnetic boundary conditions, concept of inductance.		
3	**Time Varying Fields and Maxwell's equations**		07	CO 3
	3.1	Faraday's law, transformer and motional EMF, displacement current.		
	3.2	**Maxwell's equations:** Integral and differential form for static and time varying fields and its interpretations, time harmonic fields		
4	**Electromagnetic Wave**		10	CO 4
	4.1	**Wave propagation:** Plane waves in free space, dielectrics and good conductors, Helmholtz equation, properties of plane waves and solution of wave equations.		
	4.2	**Wave parameters:** Intrinsic impedance, propagation constant, skin depth and polarization of waves		
	4.3	**Electromagnetic Power:** Poynting Vector, applications, power flow in plane wave, instantaneous, average and complex pointing vector.		
	4.4	**Wave reflections:** Reflection of plane waves at normal incidence and at oblique incidence, perpendicular and parallel		

		polarization		
5	**Applications**			
	5.1	**Applications of static fields:** Deflection of charged particles, electrostatic generator, magnetic deflection, cyclotron, Hall effect, Ink-jet printer, CRO and electromagnetic pumps.	08	CO5
	5.2	**Applications of electromagnetic fields:** Transmission lines, waveguides, Antennas, electromagnetic interference and compatibility. biological effects and safety (only construction, working, field distributions and standards)		
		Total	**45**	

Reference Books*

Sr. No.	Name/s of Author/s	Title of Book	Publisher	Edition /Year
1.	William H. Hayt	*Engineering Electromagnetics*	Tata McGraw-Hill, India	7th Edition, 2012
2.	Matthew N. O. Sadiku, S.V.Kulkarni	*Principles of Electromagnetics*	Oxford university press, India	6th Edition, 2015
3.	Bhag Guru and Huseyin Hiziroglu	*Electromagnetic Field Theory Fundamentals*	Cambridge University Press, India	2nd Edition, 2005

Instructions:	Tutorials includes journal / written record of tutorials. Tutorials will be evaluated out of **25 marks each**. A minimum of 8 tutorials to be performed in a semester.